Experiment 1: Design of Visual Memory Recall UI Using Chunking

Name: Loga Prabha GB Roll No: 240701290

Aim

To design a user interface that helps users recall visual elements such as icons or text chunks and to evaluate the effect of chunking on user memory.

Design Description

The designed UI consists of multiple screens that test the user's memory recall ability using grouped visual elements. The interface follows Human Computer Interaction principles such as simplicity, consistency, visibility, and chunking.

1. Instruction Screen

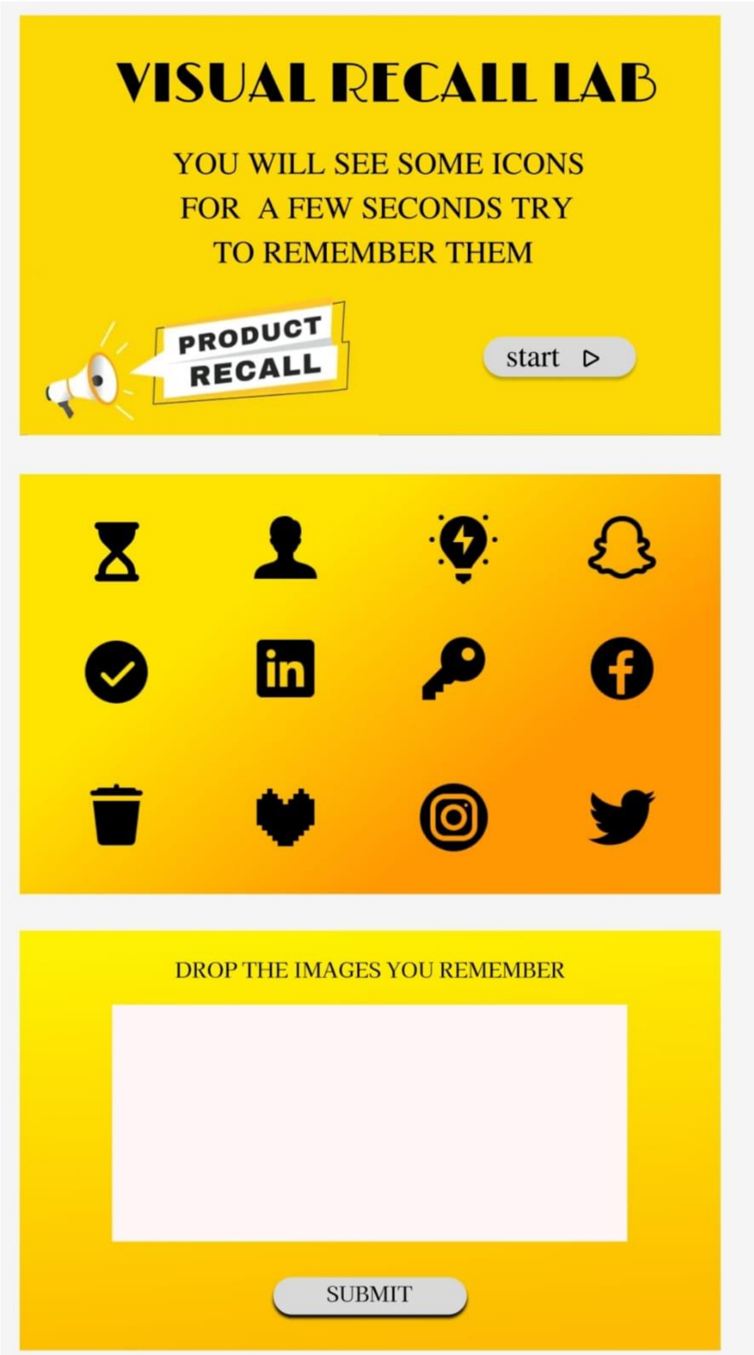
The first screen provides instructions to the user to carefully observe and remember the icons displayed. A Start button is provided to begin the experiment.

2. Observation Screen

In this screen, icons are displayed in grouped chunks rather than random arrangements. The chunking technique helps users memorize visual elements more effectively. Different sets of icons are shown for a limited time.

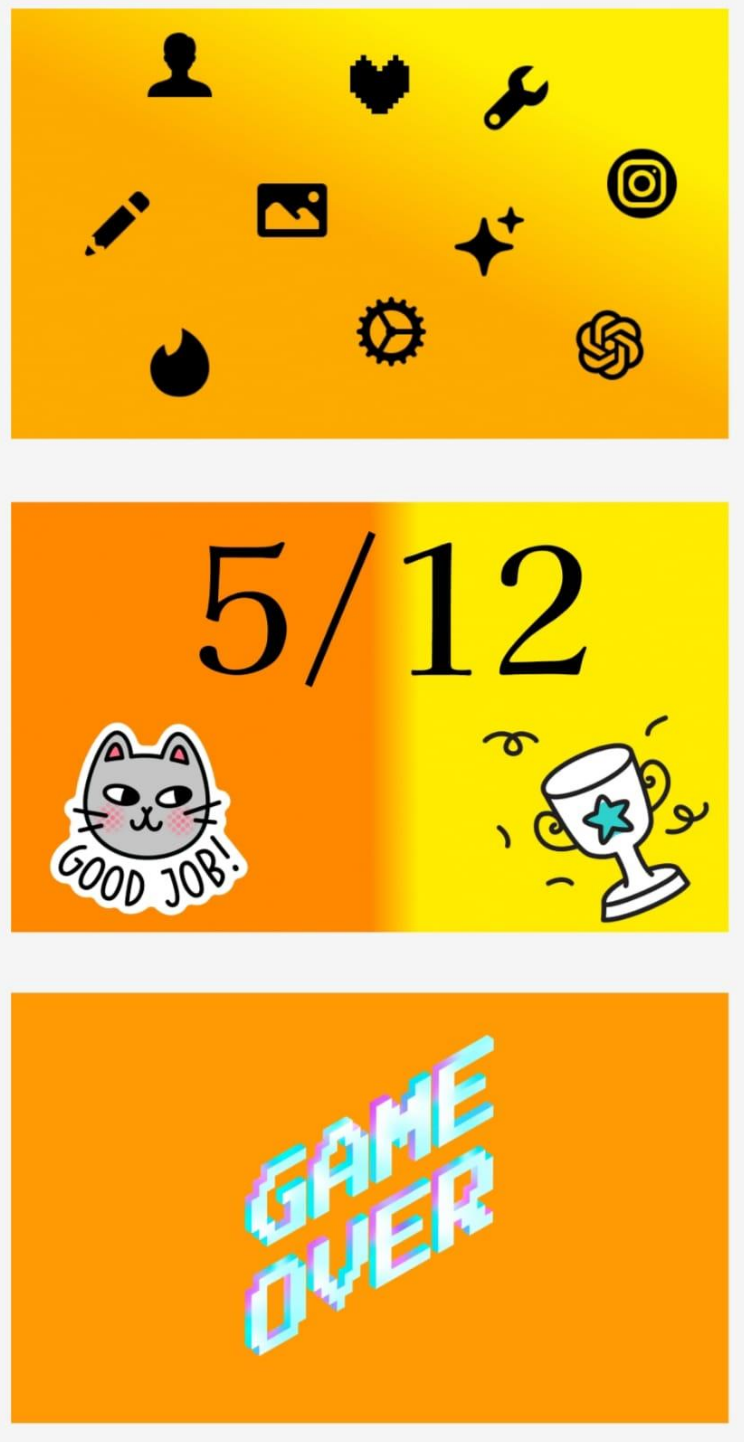
3. Recall Screen

After observing the icons, users are asked to type or identify the icons they remember. This evaluates the user's memory retention and recall accuracy.



4. Selection Screen

Users select the remembered icons from multiple options displayed on the screen. This tests recognition memory along with recall memory.



Design Features

- Soft color palette for better visual comfort
- Simple and interactive interface
- Large buttons for easy navigation
- Organized icon placement using chunking
- Minimal distractions to improve concentration

Chunking Principle Used

Chunking is the process of grouping related information together to improve memory retention. In this design:

- Icons are displayed in meaningful groups.
- Users can remember grouped items more easily.
- Cognitive load is reduced during memorization.

Working Process

1. User clicks the Start button.
2. Icons are displayed in grouped chunks.
3. User memorizes the displayed elements.
4. Recall screen appears after a few seconds.
5. User enters or selects remembered icons.
6. System evaluates recall performance.

Result

The designed UI successfully demonstrated that users could remember grouped visual elements more effectively than randomly arranged elements. Chunking improved recall accuracy and user memory performance.